Atmo-Ensemble
Ensemble ML Research Proposal for Air Quality Multi-Classification

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Abstract

The objective of this initiative is to drive a research project in order to study ensemble models and compare the best implementation ways in a multi-class problem.

When selecting the theme of study, we took into consideration the constraints stated in the project description, which mentioned that we must select imbalanced datasets in a multi-class problem with at least 4 classes in the predicted feature, staying in the tabular side of machine learning techniques. Therefore, we decided to study air quality and pollution.

This dataset focuses on air quality assessment across various regions. The dataset contains 5000 samples and captures critical environmental and demographic factors that influence pollution levels, combining concentration of principal components that influence quality of air, weather conditions, and proximity to areas of risk.

One of our reasons to select this dataset was the fact that the labeled multi-class feature contains ordinal rather than nominal information. The quality feature contains a value going from *Good* to *Hazardous*, divided into 4 different classes. These classes are correlated, as they have an order, and are not as if we classified animals or any other nominal class problem. This feature allows us to treat the data in different ways and get the best results.

Even if many interesting multi-class datasets are available on the internet fit for multi-class classification, most of them are too balanced, the objective feature contains 3 classes, is directly binomial classification, or is fully artificial. That is why after considering many options we decided to take this approach.

1 Research Proposals

The main purpose of this project is to research tabular ensemble machine learning; therefore, the objective is not just to get the best scores, but to understand the problem and draw conclusions.

1.1 PCA and Dataset Feature Extraction

Does studying correlated environmental features through principal component analysis (PCA) and other feature extraction techniques improve the robustness and efficiency of ensemble classifiers, or are non-linear feature interactions found by the models alone?

Environmental datasets inherently suffer from multicollinearity, the features are strongly correlated. In this dataset, variables as particulate matter (PM_{10} and $PM_{2.5}$) are likely to be highly correlated, just as specific industrial gases (NO_2, SO_2, CO), which might be strongly linked to the proximity to industrial areas feature.

While tree-based ensembles are generally robust to multiple correlations when making predictions, we could suffer from distortion on the metrics (which directly affects the explainability of the predictions as we will discuss in the 1.2 proposal) and add unnecessary complexity (hardening the efficiency proposal at 1.7).

Therefore, the objective is to evaluate how dimensionality reduction compares to using raw, interpretable sensor data within different ensemble architectures.

1.2 Feature Importance and Interactions (XAI)

Nowadays the understanding of the reasons for a model to make a decision are very relevant. We aim to use different techniques to interpret the models decisions. We aim to use, for instance, SHAP dependence plots to find non-linear interactions, and study the correlations of the values and each features weight in the output.

1.3 Benchmark Performance

Comparison of Bagging and Boosting

Does the variance-reduction of parallel ensembles offer better generalization on noisy sensor data compared to the bias-reduction of sequential ensembles?

We can not have both, reduced variance or bias. We will compare the two fundamental techniques of ensemble learning: reducing variance (parallel) and reducing bias (sequential).

- **Parallel (Bagging)**: Train models independently to reduce overfitting (e.g., Random Forest, Extra Trees). Good for noisy sensor data.
- **Sequential (Boosting)**: Train models iteratively to correct previous errors (e.g., XGBoost, LightGBM, CatBoost). Good for finding difficult patterns.

Therefore, the objective is to compare Random Forest (Parallel) vs. Gradient Boosting (Sequential), introducing noise to the dataset (simulating sensor error) and seeing which architecture suffers from faster metrics degradation.

1.4 Handling Ordinality

Does encoding ordinality via Regressor-based Ensembles outperform standard Multiclass Classifiers in Air Quality Assessment?

Our target variable isn't just A, B, C, D, it has a rank/order. Standard multiclass classification ignores this order (it treats "Good predicted as Hazardous" as an error equal to "Good predicted as Moderate").

We aim to study if treating the target as ordinal ($Good < Moderate < Poor < Hazardous$) in an ensemble framework improves performance over treating it as purely nominal.

The approaches consist of:

- **Standard Multiclass**: Treat classes as nominal (independent).
- **Ordinal Ensembles**:
  - *Frank and Hall Method*: Decompose the 4 classes into 3 binary classifiers (Good vs. Rest, Moderate vs. Rest, etc.) and stack them.
  - *Regressor Ensemble*: Train a Regressor (XGBoost Regressor) to predict a continuous pollution score, then map the output to the closest class.

1.5 Imbalanced Learning

1.5.1 Handling Rare Events

"Evaluating the efficacy of Hybrid Sampling-Ensemble methods (SMOTE-Boost vs. RUS-Boost) in detecting rare Hazardous Air Quality events."

Air quality data is naturally imbalanced. Hazardous samples are rare, but they are the most important to predict, as this is a recall-score critical problem, hazardous air can have lethal consecuences. Standard accuracy is not a suitable metric here (a model that always predicts *Good* might be 0.9 accurate but useless in the real world).

The main idea is to integrate sampling techniques inside the ensemble algorithms. Techniques to compare:

- **Standard Ensembles**: Random Forest, AdaBoost.
- **Imbalanced-Specialized Ensembles**:
  - *Balanced Random Forest*: Down-samples the majority class in every tree.
  - *SMOTE-Boost*: Creates synthetic minority samples (SMOTE) before each boosting iteration.
  - *RUSBoost*: Uses Random Under Sampling (RUS) during boosting.

1.5.2 Cost-Sensitive Learning vs. Resampling Methods

A parallel approach to data resampling is algorithmic modification via Cost-Sensitive Learning. Instead of changing the data, we could change the loss function of the algorithm to heavily penalize dangerous mistakes. Because our target is ordinal, predicting "Hazardous" as "Good" is objectively worse than predicting "Hazardous" as "Poor". We will implement a custom loss function using a penalty matrix.

$$L(y, \hat{y}) = \sum_{i=1}^{N} C(y_i, \hat{y}_i)\,(y_i - \hat{y}_i)^2$$

1.5.3 Robustness to Missing Data (Simulating Broken Sensors)

Air quality sensors fail constantly in the real world, and even if our dataset looks clean, we would like to train solutions that can handle real life situations. Apart from adding noise, we will simulate data dropout. We will randomly drop 0.1, 0.2, or 0.3 of the

$PM_{2.5}$ or NO_2 records and compare two approaches: using a standard ensemble combined with an imputation technique (like k-NearestNeighbour or mean imputation) versus using XGBoost or LightGBM, which have native, built-in algorithms for handling NaN values.

1.6 Semi-Supervised Learning

Most times, manually labeling a complete dataset can be very time consuming, and in real life scenarios, time is money. We are aiming to partially delete the target feature in different proportions for each class, and run active and pasive semi-supervised learning processes. This experiment will help us understand how it performs with our samples. Our set of samples is not as large as we would like it to be, but it will be useful to study how to identify the most interesting samples to manually label and leave the rest to the model, saving the time and effort it takes to fully supervise a dataset.

1.7 Efficiency vs. Accuracy

1.7.1 Trade-offs between Computational Cost and Classification Performance

Can lightweight ensembles replace deep stacking for edge-based Air Quality Monitoring?

Edge machine learning is a heavily discussed topic. Air quality monitoring is often done on edge devices (small, low-power sensors in the city), and running a massive stacking ensemble might be too heavy for a battery-powered or low power chip. We aim to find the frontier between model size and performance.

- **Heavy Model**: A 3-layer Stacking Classifier (XGBoost + RF + SVM).
- **Light Model**: A highly pruned Decision Tree or a LightGBM with limited depth.
- **Measurement**: Measure not just Accuracy/F1-Score, but also Inference Time (ms) and Model Size (MB).

1.7.2 Conformal Prediction

Instead of forcing the model to output a single prediction (e.g., "Poor"), conformal prediction allows the model to output a set of predictions with a defined chance of containing the true class (e.g., predicting {*Moderate, Poor*} with 0.95 confidence).

We will study ways of changing the size of the prediction set between an uncalibrated Random Forest and a well-calibrated LightGBM and how it reacts when faced with noisy sensor data.

Therefore, we look into the drawbacks of modifying the loss function in order to heavily penalize hazardous quality miss-classifications and understanding if ordinal costs outperform synthetic data generation.

2 Dataset Feature Analysis

The features we are going to work with are the following:

- **Temperature** (°C): Average temperature of the region.
- **Humidity** (%): Relative humidity recorded in the region.
- $PM_{2.5}$ **Concentration** ($\mu g/m^3$): Fine particulate matter levels.
- PM_{10} **Concentration** ($\mu g/m^3$): Coarse particulate matter levels.
- NO_2 **Concentration** (ppb): Nitrogen dioxide levels.
- SO_2 **Concentration** (ppb): Sulfur dioxide levels.
- CO **Concentration** (ppm): Carbon monoxide levels.
- **Proximity to Industrial Areas** (km): Distance to the nearest industrial zone.
- **Population Density** (people/km^2): Number of people per square kilometer in the region.

Target Variable: Air Quality Levels

- **Good**: Clean air with low pollution levels.
- **Moderate**: Acceptable air quality but with some pollutants present.
- **Poor**: Noticeable pollution that may cause health issues for sensitive groups.
- **Hazardous**: Highly polluted air posing serious health risks to the population.

3 State of the Art and Previous Approaches

Most community members used tabular machine learning models individually, including Logistic Regression, Decision Trees, Naive Bayes, Support Vector Machines (SVM), GBM, Extreme Gradient Boosting (XGBoost), and MLP, achieving best results with Decision Trees (0.95+ F1 score). Even if Random Forests represent a consensus model of many decision trees, we aim to approach the problem using ensemble models in different ways, comparing techniques to gain insights into our research objectives.